



CESAB
CENTRE DE SYNTHÈSE ET D'ANALYSE
SUR LA BIODIVERSITÉ

Pooling effect sizes

Common-effect, random-effects and
multilevel models — and how to show the
results

Damien Beillouin
CIRAD, UPR HortSys

SOUTIENS INSTITUTIONNELS



Learning objectives

By the end of this session, you will be able to:

- 1 Explain why studies are **weighted**, and how the weights change between models.
- 2 Distinguish **common-effect**, **random-effects** and **multilevel** models – and choose one **before** seeing the data.
- 3 Read μ , its CI, τ^2 , I^2 and the **prediction interval**, and say which answers which question.
- 4 Fit and plot them in R with **metafor** (practical, 33 min).

Where we are

This morning, `escalc()` gave us, for each of the **382** comparisons of pest abundance:

- y_i : the lnRR (effect size)
- v_i : its sampling variance

These come from **28 studies**.

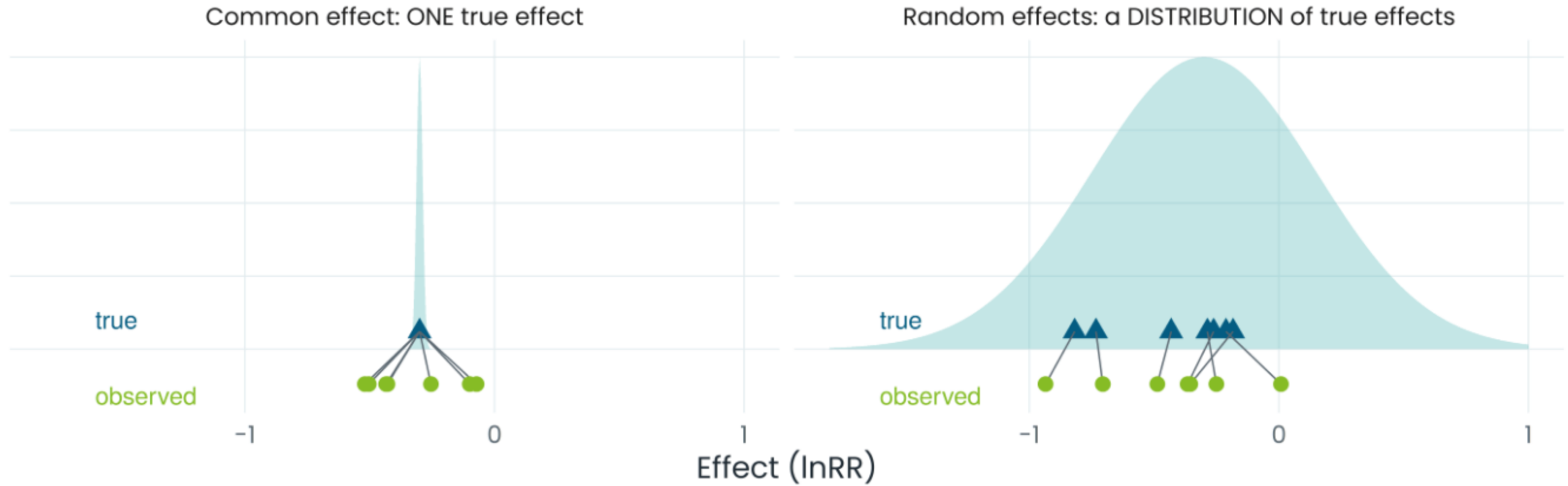
This afternoon's question

How do we turn 382 numbers into **one answer** – and an honest statement of **how certain** and **how variable** it is?

Plain average of the 382 lnRR: **-24%**. Why is that a bad answer? (*think 20 s*)

Weights and models

Two views of the world



$y_k = \mu + \varepsilon_k$ — studies differ **only** by sampling error.

Also called “fixed-effect” or “equal-effects” (method = “EE”).

$y_k = \mu + u_k + \varepsilon_k$, with $\text{Var}(u_k) = \tau^2$ — true effects **really differ** (soils, climates, pests...).

Weights: precise studies count more

Common effect:

$$w_k = \frac{1}{v_k}$$

Random effects:

$$w_k = \frac{1}{v_k + \tau^2}$$

Adding τ^2 to every variance makes the weights **more equal**: very precise studies dominate less.

Share of the total weight	Common effect	Random effects
Zhou 2013 (51 very precise rows)	76%	17%
Maluleke 2005 (80 rows)	23%	26%

Common effect: **2 studies out of 28 carry 99%** of the weight.
Random effects: τ^2 tames the *precise* studies — but a study with **many rows** still counts a lot. That is the next problem.

Which model? Decide before seeing the data

SHOW OF HANDS

Q is significant ($p < 0.001$). Should I switch from a common-effect to a random-effects model?

No — the choice is conceptual, not a test result. Do you want to generalise beyond these exact studies? Do you expect true effects to differ (sites, species, practices)? In ecology and agronomy: **yes and yes → random effects, stated in the protocol.** The Q test has low power with few studies and is almost always significant with many.

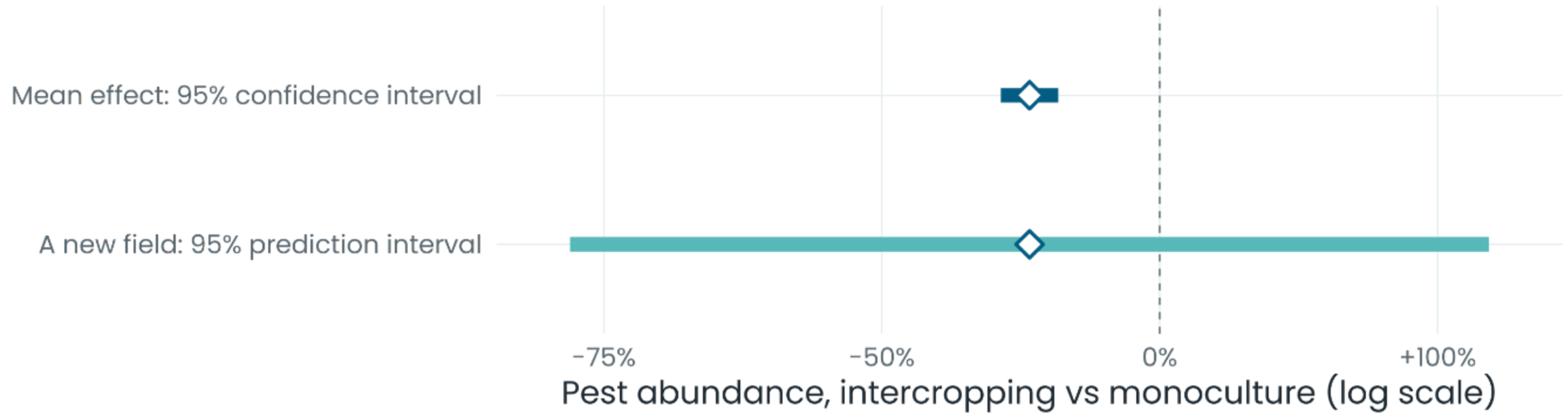
Heterogeneity

Three numbers, three meanings

	Meaning	Our pests (random-effects model, dependence ignored for now)
τ^2 (τ)	Variance (SD) of the true effects, in lnRR units	$\tau = 0.58$
I^2	Share of the observed variance not due to sampling error — a <i>relative</i> measure	99.9%
Q	Test of “all true effects are equal”	$p < 0.001$

$I^2 \approx 100\%$ does not mean “huge differences”: it means sampling error is tiny compared with them — typical when studies are large. I^2 says nothing about **how different** the effects are. For that, use τ and the **prediction interval**.

Confidence vs prediction interval



CI: -33% to -22% — how precisely we know the **average**. Shrinks as studies accumulate.

PI: -77% to +128% — where the effect in a **new** field is expected. Reflects τ^2 ; does not shrink (IntHout et al. 2016).

Quick quiz

1. $I^2 = 0\%$. Are the true effects identical?

Not necessarily: with few, imprecise studies, heterogeneity is simply **not detectable**.

2. A farmer asks: “will intercropping reduce pests **on my farm?**” CI or PI?

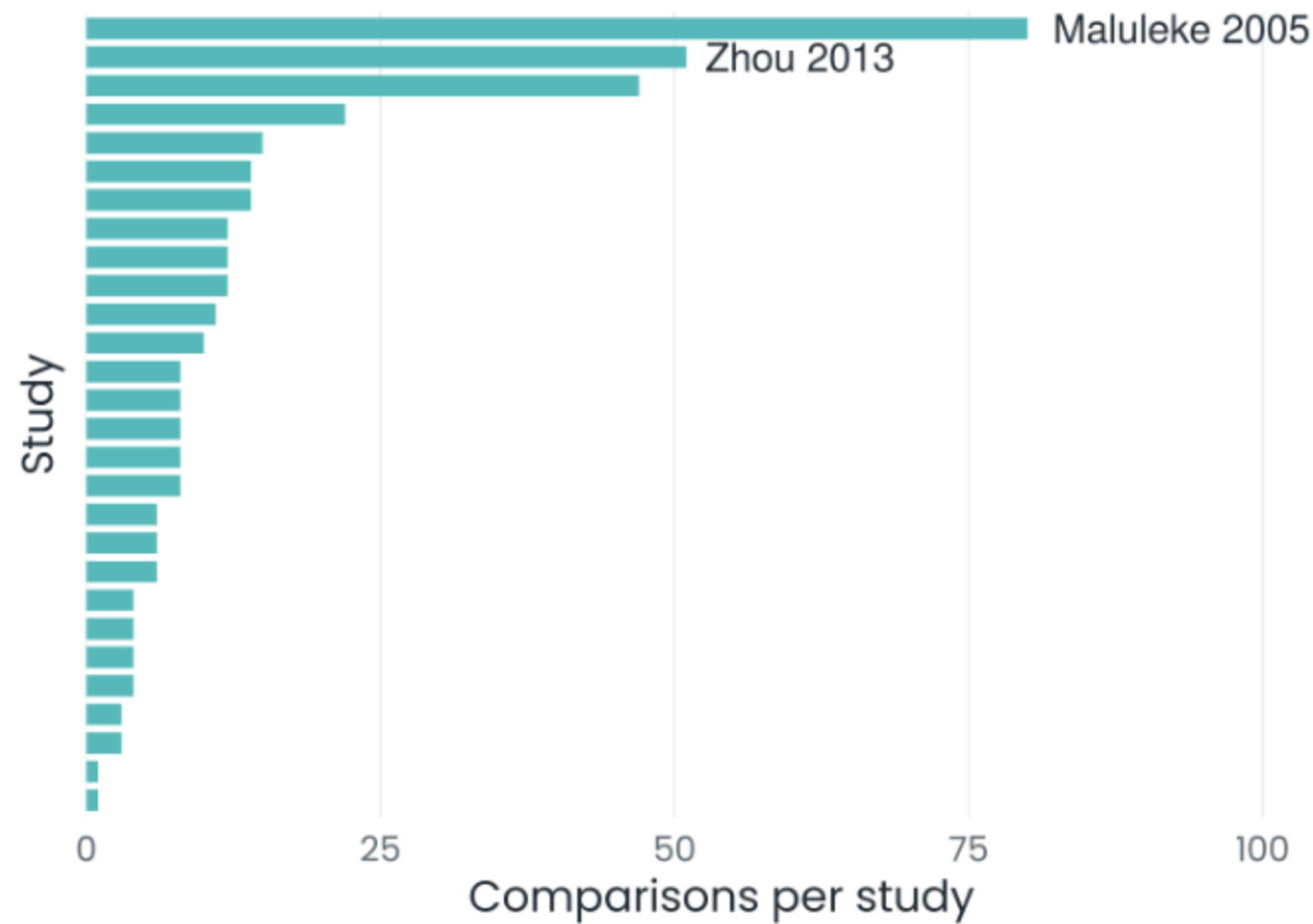
PI — and here it includes increases.

3. $\tau = 0.58$ on the lnRR scale. Small or large?

Large: mean $\pm 1 \tau$ already spans -60% to +29%.

Non- independence

382 comparisons are not 382 studies



Several taxa, dates, sites or treatments **within the same study**: same team, same field, same weather, sometimes the **same control plots**.

Treating them as independent = **pseudo-replication**: CIs too narrow, studies with many comparisons over-weighted.

The multilevel (three-level) model

$$y_{ij} = \mu + u_j + u_{ij} + \varepsilon_{ij}$$

- u_j : **between-study** variation, σ_{study}^2
- u_{ij} : **within-study** variation (between comparisons), σ_{within}^2
- ε_{ij} : sampling error, v_{ij} (known)

study_id / es_id = rows **nested** in studies:

```
1 rma.mv(yi, vi,  
2       random = ~ 1 | study_id / es_id,  
3       test = "t", dfs = "contain",  
4       data = dat)
```

Our pests

Of the total variance, **52%** is **between studies** and **48% within studies** (multilevel I^2 , Nakagawa and Santos (2012)):

$$I_{study}^2 = \frac{\sigma_{study}^2}{\sigma_{study}^2 + \sigma_{within}^2 + \tilde{v}}$$

$\tilde{v}$ = a "typical" sampling variance.

test = "t", dfs = "contain": t-based CIs whose degrees of freedom follow the number of **studies** ($28 - 1 = 27$), not of comparisons (381) — safer than z.

Predict first

SHOW OF HANDS

Same 382 comparisons, four models. Which one gives the widest confidence interval — and does the mean change much?

Same data, four models



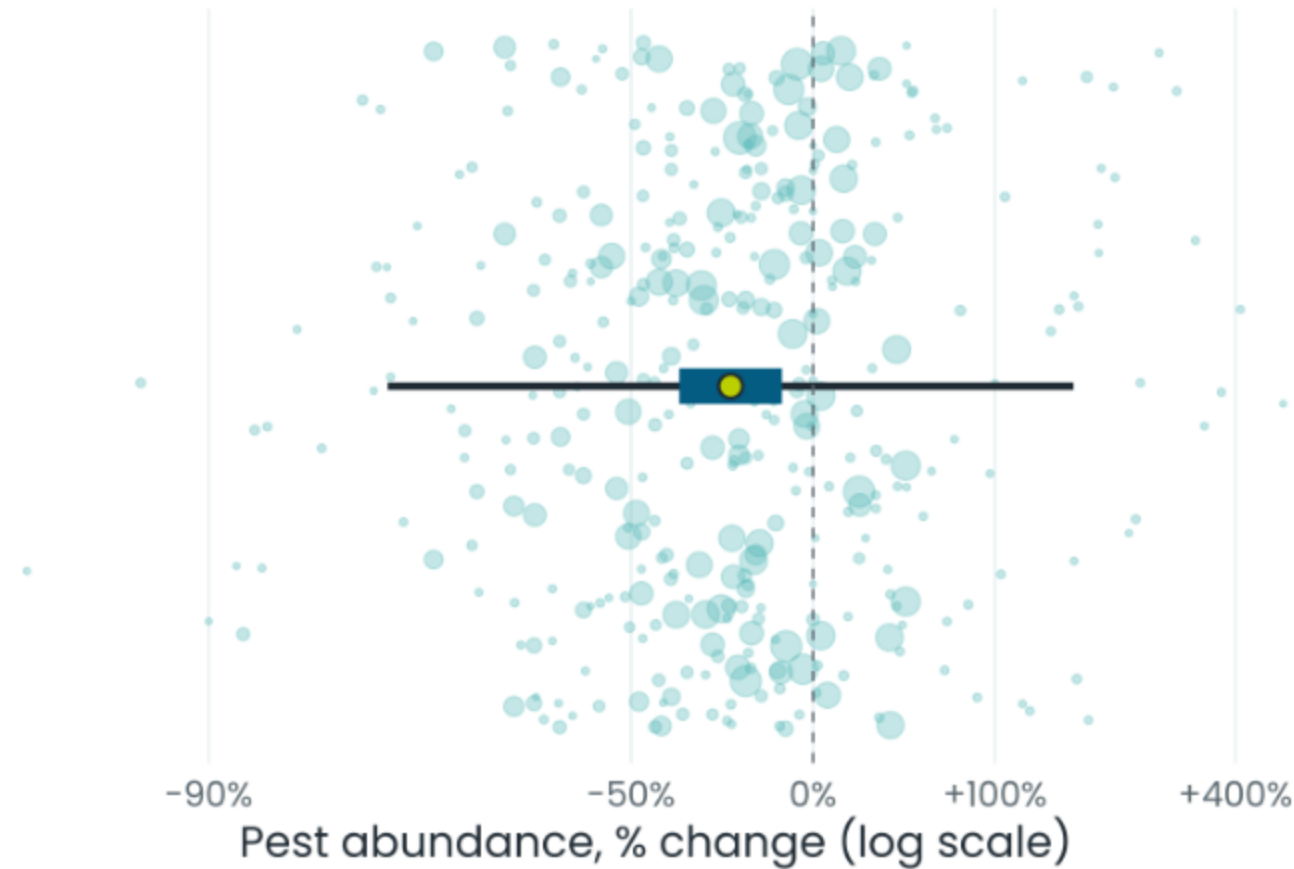
The first two models are **overconfident**: one study dominates (common effect) or 382 comparisons are treated as 382 studies.

Robust variance estimation (`robust()`, clustered by study) is a safety net when the dependence structure is uncertain (Pustejovsky and Tipton 2022). Needs ≥ 10 –20 studies.

Showing the results

Beyond the forest plot: the orchard plot

k = 382 comparisons, 28 studies



- **Points:** every comparison (y_i), sized by precision
- **Thick bar:** 95% CI of the mean — $ci.lb, ci.ub$
- **Thin bar:** 95% prediction interval — $pi.lb, pi.ub$
- **Dot:** the mean — $pred$ (all from $predict(model)$)
- **k** and number of studies

With hundreds of effect sizes, a classic forest plot is unreadable; the orchard plot shows **mean, uncertainty, heterogeneity and raw data** at once (Nakagawa et al. 2021).

Practical · 33 min + debrief

Time	Task
14:22	Common vs random effects: estimates and weights
14:30	Heterogeneity: τ^2 , I^2 , prediction interval
14:38	Multilevel model + <code>robust()</code>
14:46	Orchard plot , then <i>your turn</i> : change one word, rerun for natural enemies
14:55	Debrief (all together)

Open `meta_analysis_course.Rproj` (the course kit), then `td_models.R`. Instructions and folded solutions: the practical web page.

KEY MESSAGES

- 1 | Studies are **weighted** by precision; τ^2 (random effects) tames very precise studies, the **multilevel** model stops studies with many rows from dominating.
- 2 | Choose the model **a priori**; in ecology: **random effects**, and **multilevel** when a study gives several effect sizes.
- 3 | Report **μ with its CI, τ^2 and the prediction interval**; I^2 is relative, not a measure of how different effects are.
- 4 | Show the data: **orchard plots**. *Next (15:00): are these 28 studies a biased sample?*



In the online notebook

Each point of this session developed further, with the same data, more pitfalls and exercises with solutions:

- Dependence: map it before you model it
- Combining effect sizes: fixed, random and multilevel models
- One question from start to finish

literaturesynthesis.github.io/notebook

References

- Borenstein, M., L. V. Hedges, J. P. T. Higgins, and H. R. Rothstein. 2021. Introduction to meta-analysis. Second edition. Wiley, Chichester.
- IntHout, J., J. P. A. Ioannidis, M. M. Rovers, and J. J. Goeman. 2016. [Plea for routinely presenting prediction intervals in meta-analysis](#). BMJ Open 6:e010247.
- Jones, S. K., A. C. Sánchez, S. D. Juventia, and N. Estrada-Carmona. 2021. [A global database of diversified farming effects on biodiversity and yield](#). Scientific Data 8:212.
- Maluleke, M. H., A. Addo-Bediako, and K. K. Ayisi. 2005. [Influence of maize/lablab intercropping on lepidopterous stem borer infestation in maize](#). Journal of Economic Entomology 98:384–388.
- Nakagawa, S., M. Lagisz, R. E. O'Dea, J. Rutkowska, Y. Yang, D. W. A. Noble, and A. M. Senior. 2021. [The orchard plot: Cultivating a forest plot for use in ecology, evolution, and beyond](#). Research Synthesis Methods 12:4–12.
- Nakagawa, S., and E. S. A. Santos. 2012. [Methodological issues and advances in biological meta-analysis](#). Evolutionary Ecology 26:1253–1274.
- Nakagawa, S., Y. Yang, E. L. Macartney, R. Spake, and M. Lagisz. 2023. [Quantitative evidence synthesis: A practical guide on meta-analysis, meta-regression, and publication bias tests for environmental sciences](#). Environmental Evidence 12:8.
- Pustejovsky, J. E., and E. Tipton. 2022. [Meta-analysis with robust variance estimation: Expanding the range of working models](#). Prevention Science 23:425–438.
- Rice, K., J. P. T. Higgins, and T. Lumley. 2018. [A re-evaluation of fixed effect\(s\) meta-analysis](#). Journal of the Royal Statistical Society: Series A 181:205–227.
- Senior, A. M., C. E. Grueber, T. Kamiya, M. Lagisz, K. O'Dwyer, E. S. A. Santos, and S. Nakagawa. 2016. [Heterogeneity in ecological and evolutionary meta-analyses: Its magnitude and implications](#). Ecology 97:3293–3299.
- Zhou, H.-B., J.-L. Chen, Y. Liu, F. Francis, E. Haubruge, C. Bragard, J.-R. Sun, and D.-F. Cheng. 2013. [Influence of garlic intercropping or active emitted volatiles in releasers on aphid and related beneficial in wheat fields in China](#). Journal of Integrative Agriculture 12.